Computer Vision

Segmentation using CNN

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School of Computing

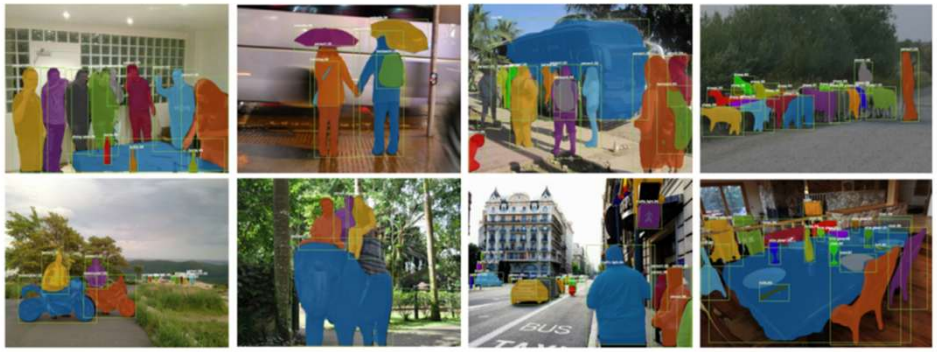
가천대학교 AI·소프트웨어학부
Gachon University

Image Segmentation using CNN

Image Segmentation

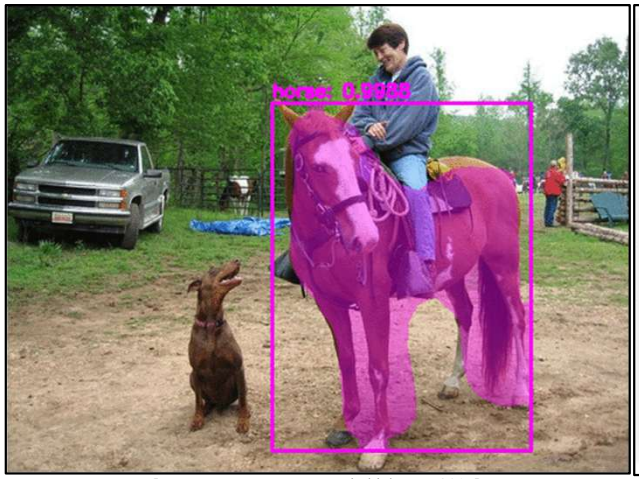
- Image segmentation divides an image into meaningful regions at the pixel level.
- Unlike object detection, which uses bounding boxes, image segmentation identifies the pixels belonging to each region or object.

- **Semantic Segmentation**: Assigns a class label to every pixel without distinguishing individual objects of the same class.
- **Instance Segmentation**: Identifies and segments each individual object separately.
- **Panoptic Segmentation**: Combines semantic and instance segmentation, assigning a class label to every pixel and distinguishing individual object instances.



Segmentation

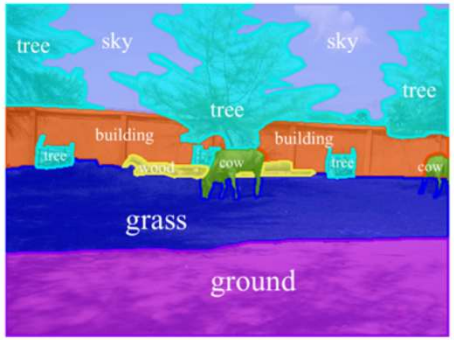


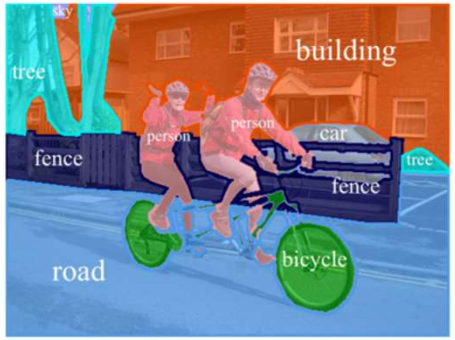


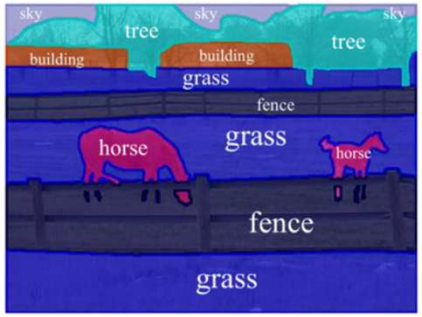
[Faster R-CNN: Ren, He, Girshick, Sun 2015]

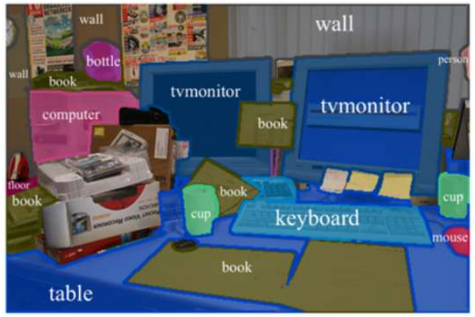


[Semantic segmentation with OpenCV and deep learning: A. Rosebrock 2018]



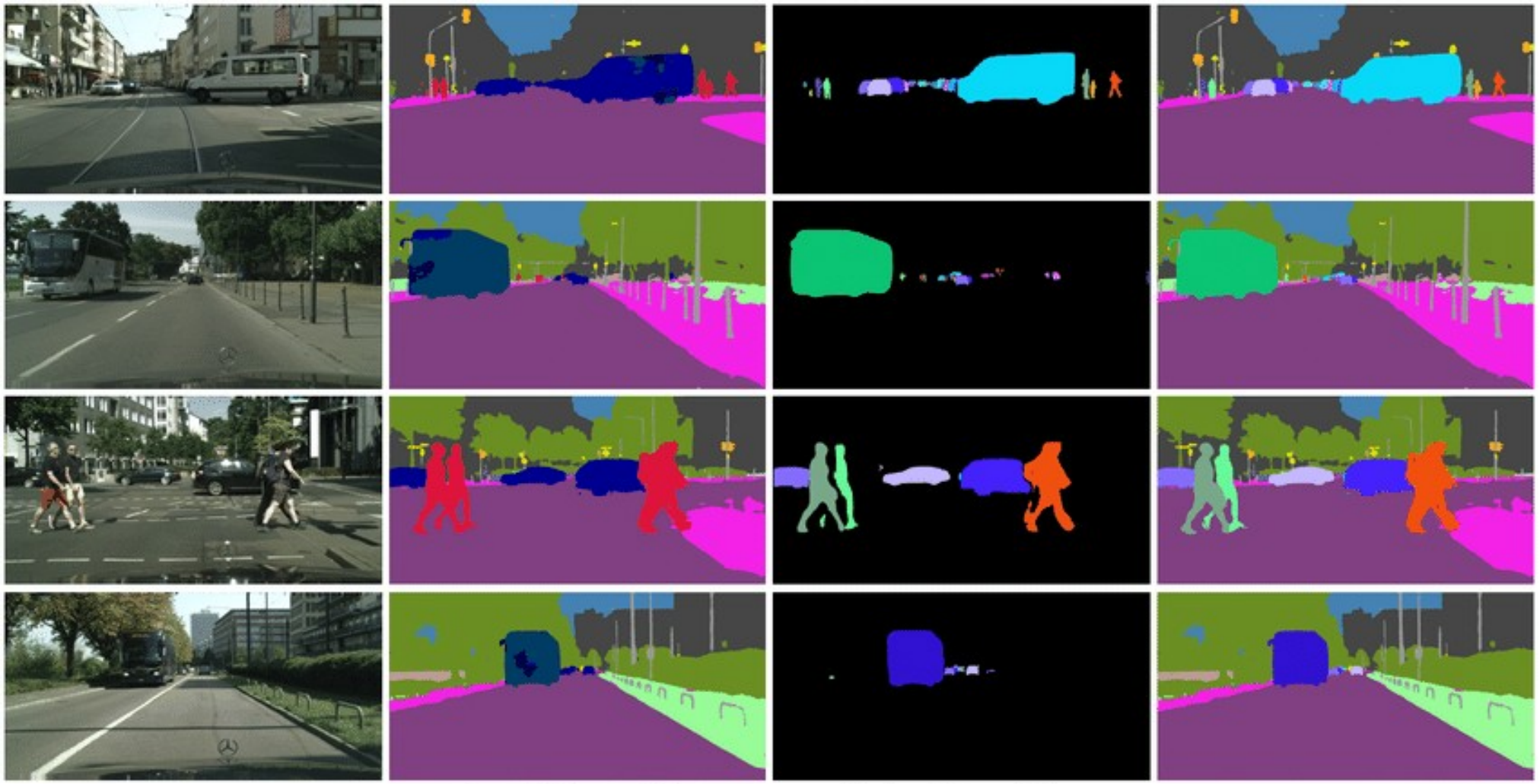








Segmentation



A. D. Cistea, et al. “Fusion Scheme for Semantic and Instance-level Segmentation,” 2018 IEEE International Conference on Intelligent Transportation System

Segmentation

- Self-driving cars using semantic segmentation

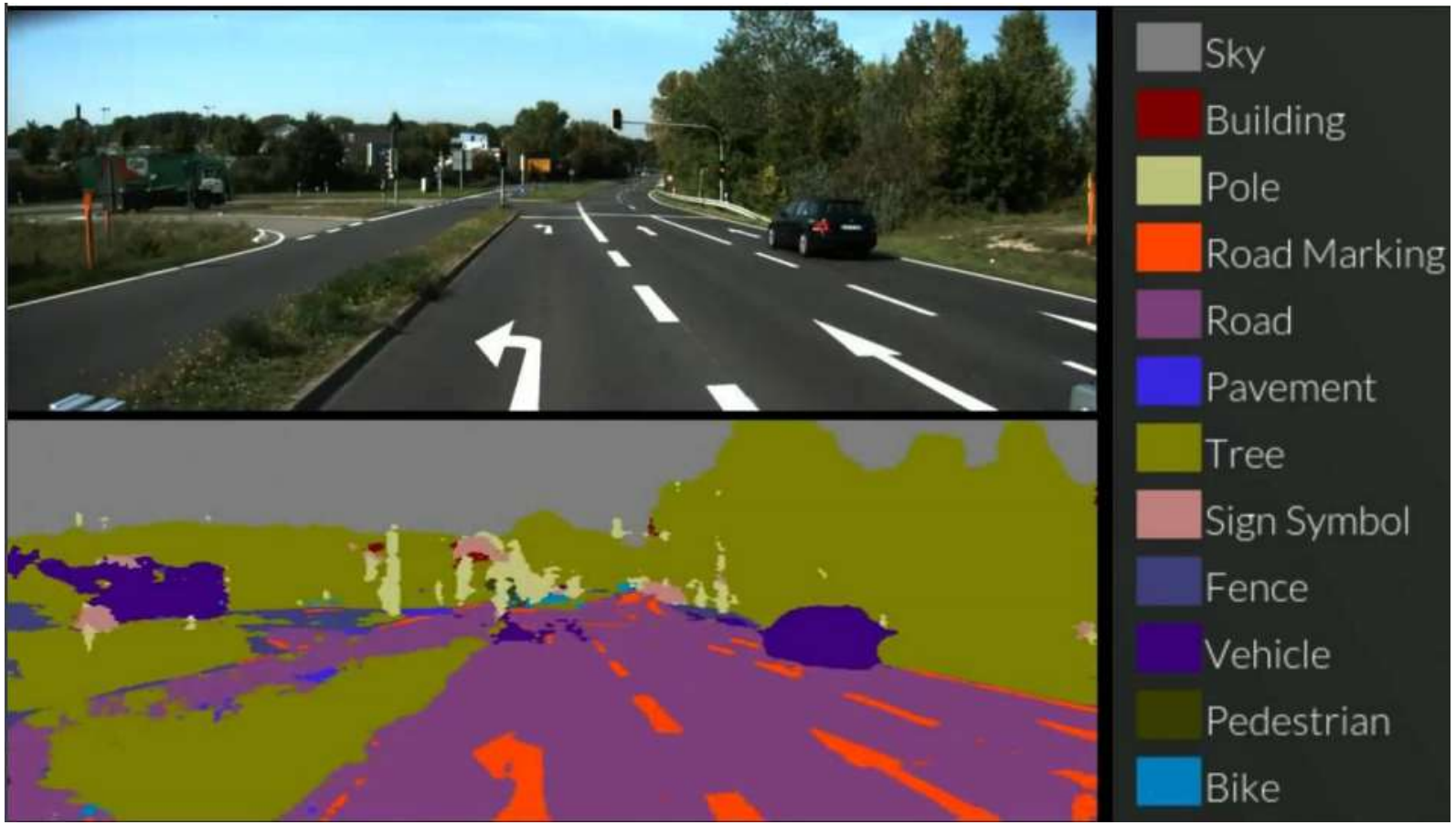


Image source: https://chatbotslife.com/small-u-net-for-vehicle-detection-9eec216f9fd6

Panoptic Segmentation

- Self-driving cars using panoptic segmentation

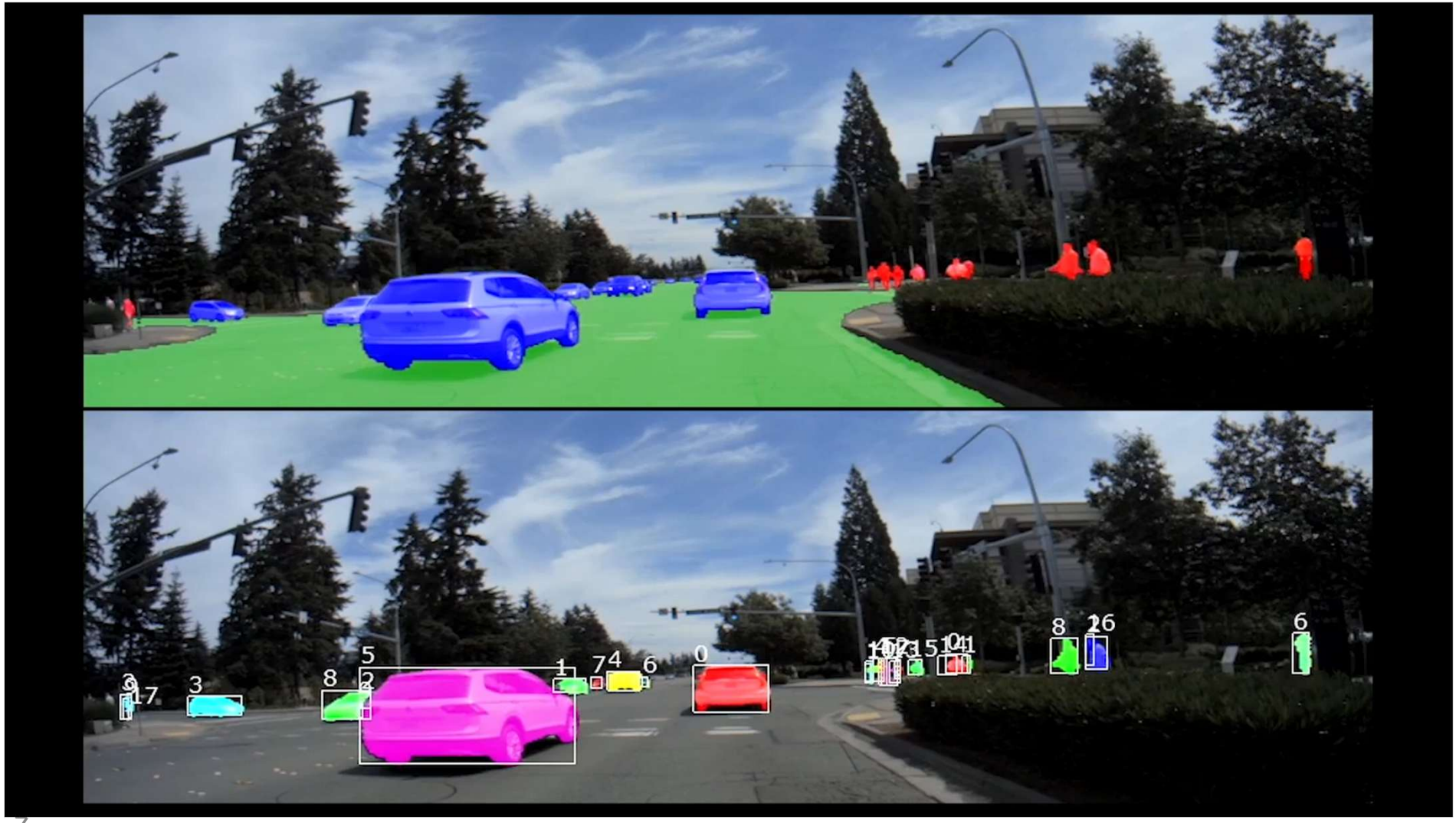
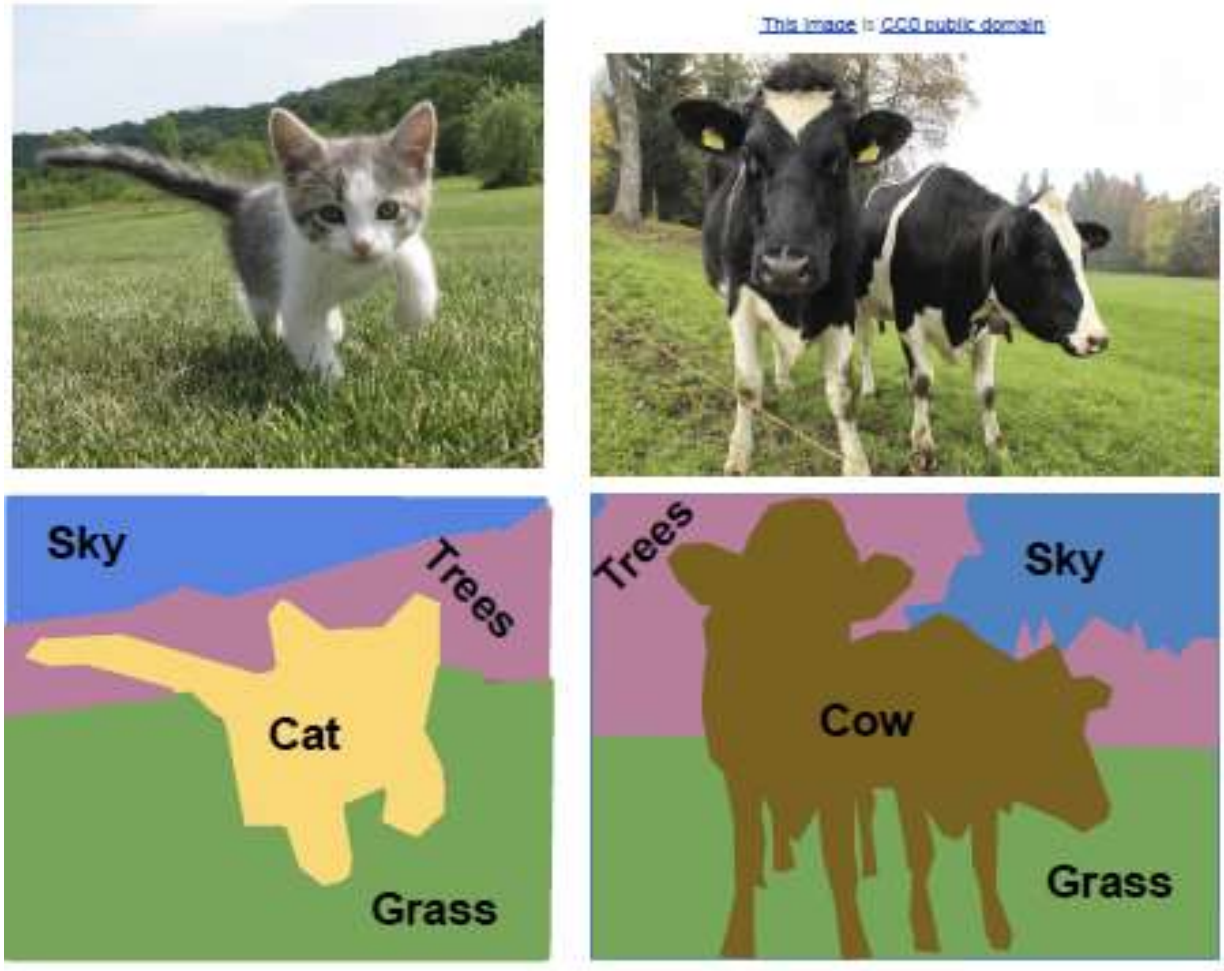


Image source: https://blogs.nvidia.com/blog/2019/10/23/drive-labs-panoptic-segmentation/

Semantic Segmentation

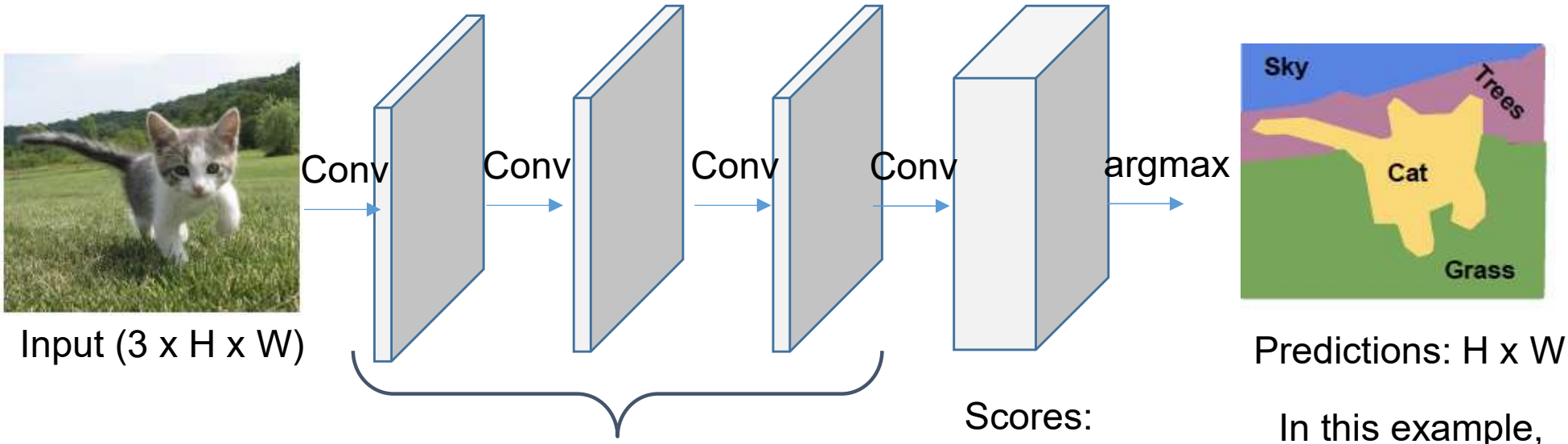
- Label each pixel in the image with a category label



Slide: cs231n, Stanford Univ. 2021

Fully Convolutional Segmentation Network

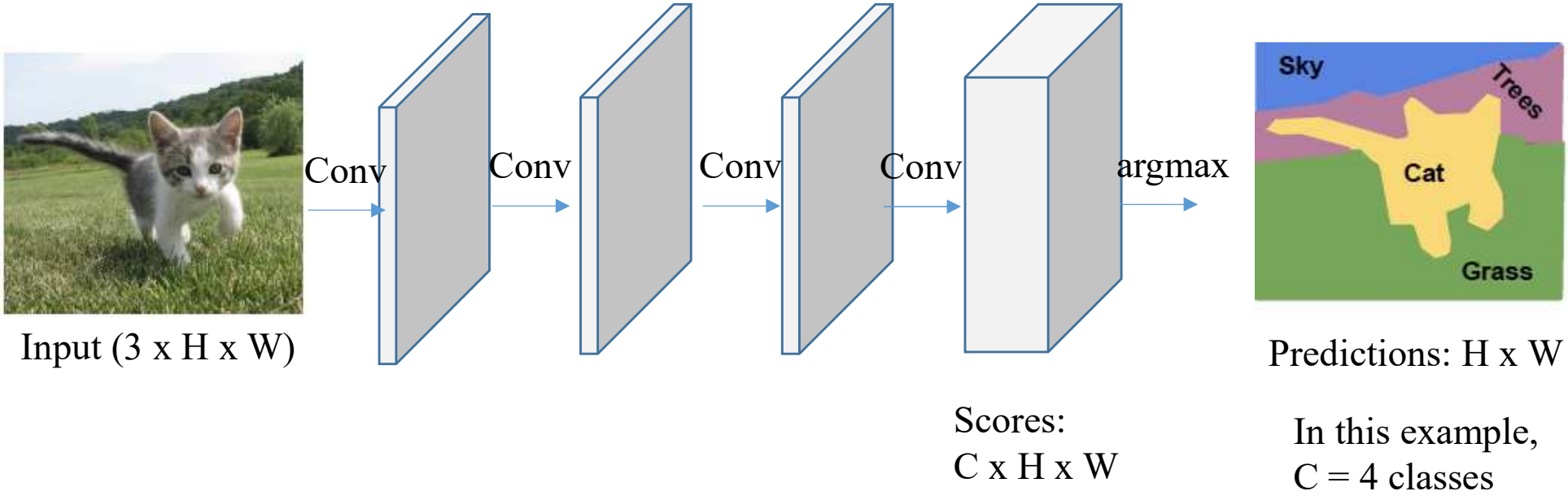
- We can design a network with fully convolutional layers.
- The final layer is to output the per-pixel scores for C classes and then we can take only the class label with the maximum score for each pixel.
- Trained with ground-truth segment map and using **cross-entropy loss** because this is a per-pixel classification problem.



Slide: cs231n, Stanford Univ. 2021

Fully Convolutional Segmentation Network

- Design a network as a bunch of convolutional layers to make predictions for pixels all at once!

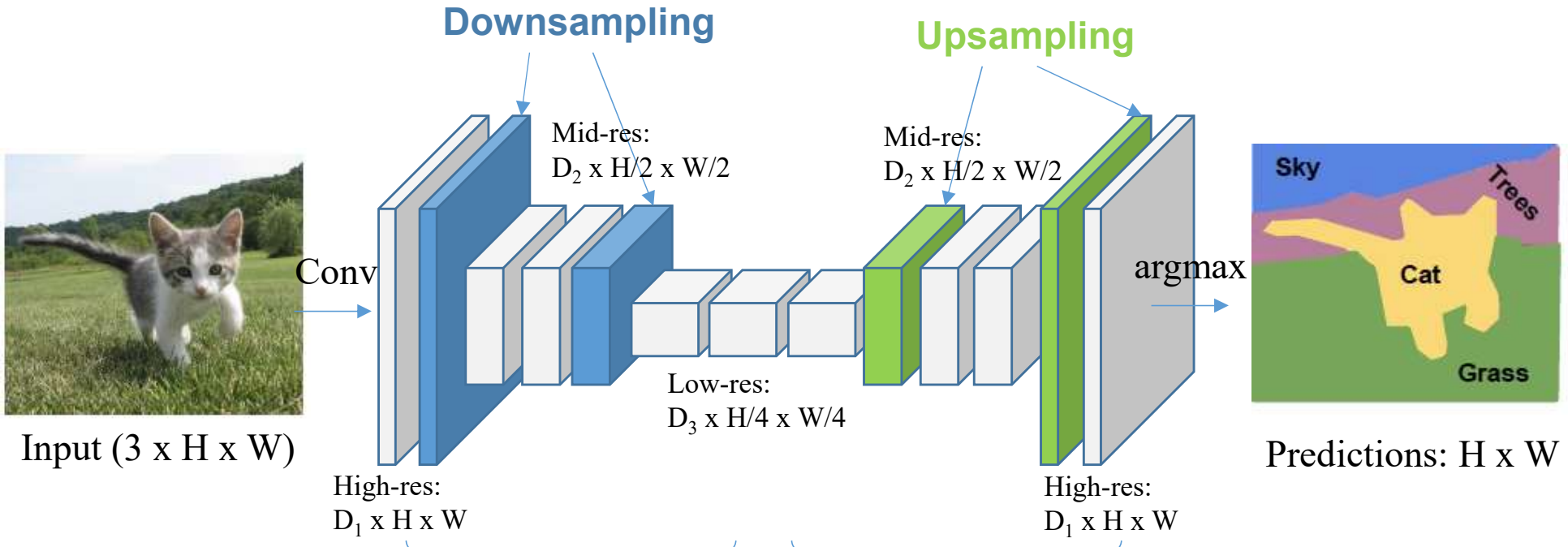


Problem: convolutions at original image resolution will be very expensive

Slide: cs231n, Stanford Univ. 2021

Fully Convolutional Segmentation Network

- **Encoder-Decoder network** is better than the previous version.
  - The network consists of convolutional layers with **downsampling** and **upsampling** inside the network.
- Trained with ground-truth segment map and using **cross-entropy loss**



Slide: cs231n, Stanford Univ. 2021

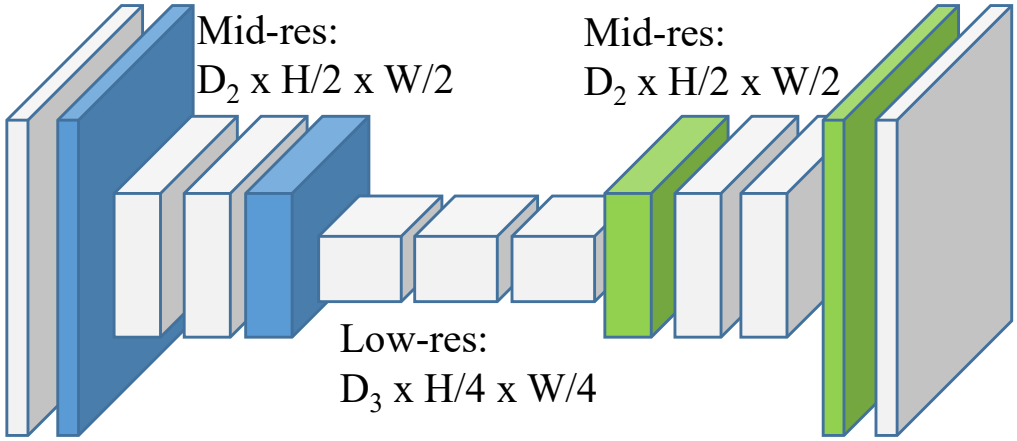
Encoder-Decoder Network

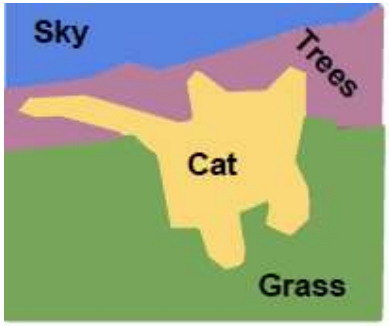
Downsampling:
Pooling, strided convolution

Upsampling:
Strided transposed convolution or Simple interpolation



Input:
3 x H x W





Predictions:
H x W

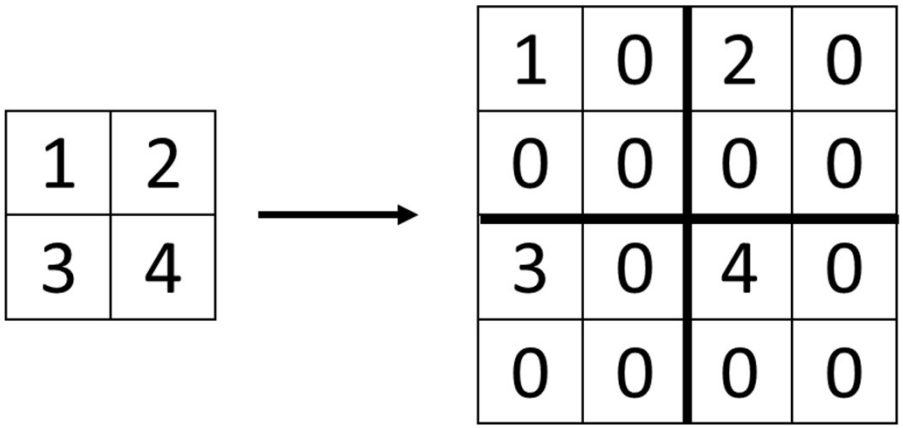
Encoder:
Contraction part

Decoder:
Expansion part

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In-Network Upsampling: “Unpooling”

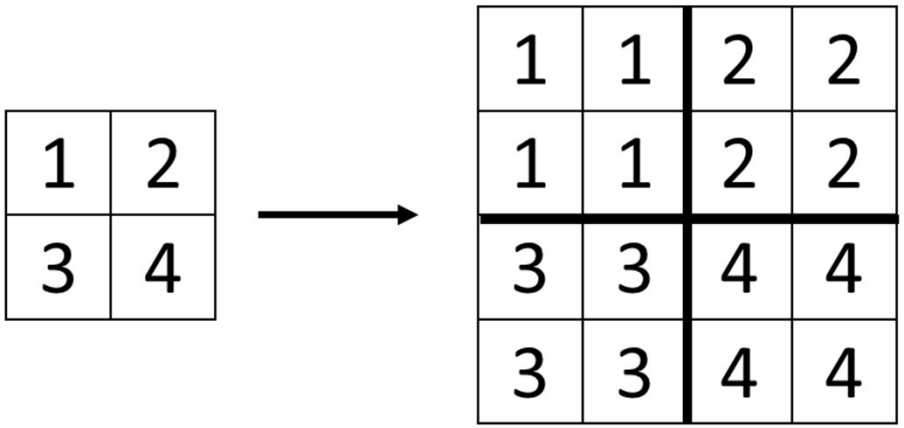
Bed of Nails



Input
C x 2 x 2

Output
C x 4 x 4

Nearest Neighbor



Input
C x 2 x 2

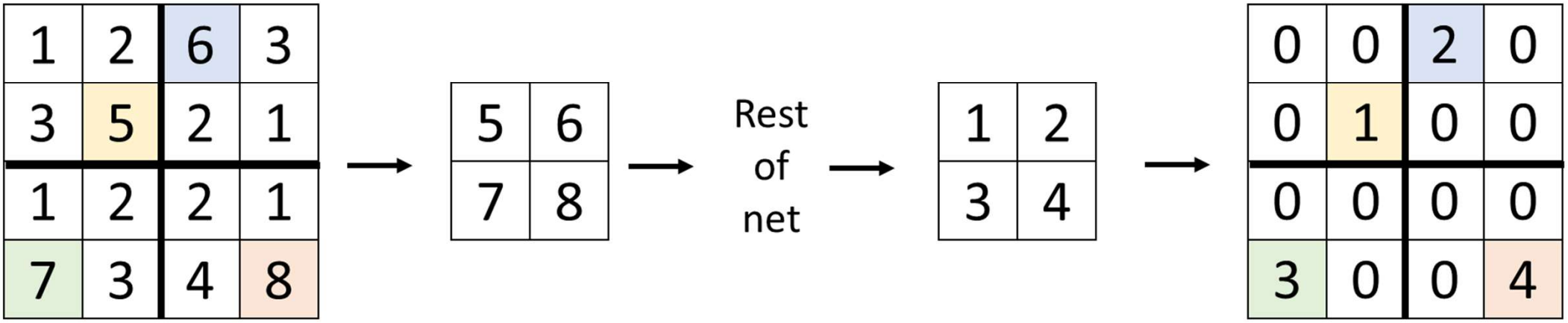
Output
C x 4 x 4

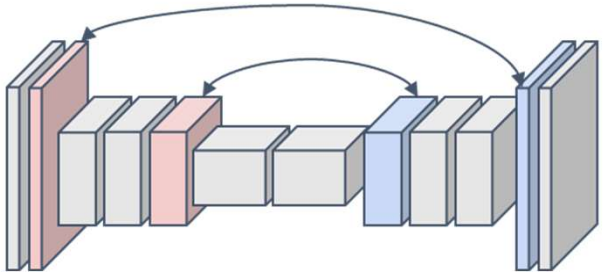
Szeliski, Chapter 6.3

In-Network Upsampling: Max Unpooling

Max Pooling: Remember which position had the max

Max Unpooling: Place into remembered positions

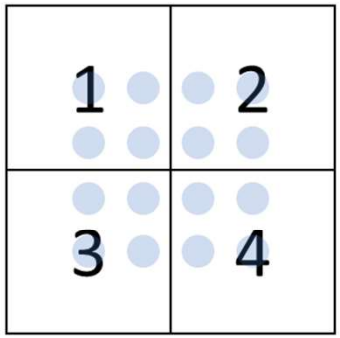




Pair each downsampling layer with an upsampling layer

Szeliski, Chapter 6.3

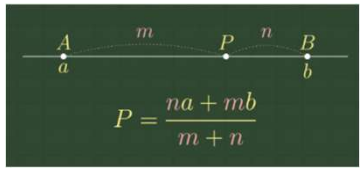
In-Network Upsampling: Bilinear Interpolation

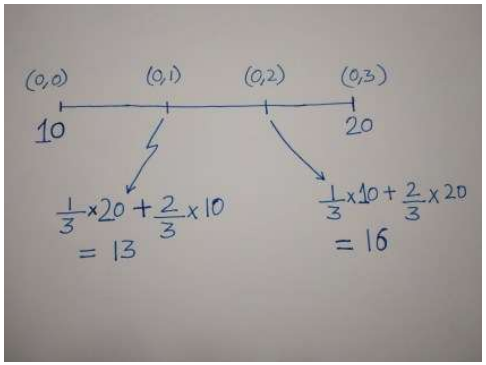


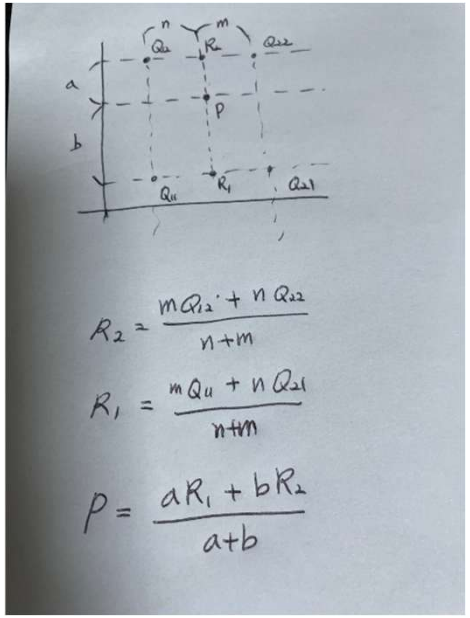
1.00	1.25	1.75	2.00
1.50	1.75	2.25	2.50
2.50	2.75	3.25	3.50
3.00	3.25	3.75	4.00

Input: C x 2 x 2

Output: C x 4 x 4







Szeliski, Chapter 6.3